Imperial College London
Business School
Machine Learning and Finance Course

Coursework Project

Dividend-Signal Alpha: Linking Earnings Surprises to Dividend Changes for a Dollar-Neutral Equity Strategy

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Group Size: 2–3 students

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1. Introduction and Motivation

This project investigates a central but under-exploited link in equity markets: the informational channel running from earnings announcements to subsequent dividend decisions, and the medium-term return dynamics that follow. You will build a complete quantitative research pipeline — from raw data to a fully backtested dollar-neutral trading strategy — that exploits the lag between earnings information and the market's repricing of dividend expectations.

Why dividends matter beyond income. Dividend changes are among the strongest managerial signals available to equity investors. Unlike earnings, which are subject to accounting discretion, a dividend commitment is a hard cash obligation. Classic signaling models (Bhattacharya 1979, Miller & Rock 1985) predict that dividend increases convey management's confidence in sustainable cash flows, while cuts signal inability to maintain the payout. Empirical evidence confirms that dividend changes carry incremental information content and trigger significant price reactions (Aharony & Swary 1980, Michaely, Thaler & Womack 1995). However, the nature of that information is debated: Grullon, Michaely & Swaminathan (2002) show that dividend-increasing firms subsequently experience declining profitability and investment, suggesting that increases may signal maturity and declining growth rather than improving prospects. Benartzi, Michaely & Thaler (1997) find only limited support for the prediction that dividend changes forecast future earnings. What is not in dispute is that the market reacts strongly to dividend changes and that this reaction unfolds over weeks to months — which is the exploitable phenomenon at the heart of this project.

The earnings-to-dividend information lag. Markets react to earnings surprises within hours. However, the second-order implication of an earnings surprise for the upcoming dividend decision propagates much more slowly. Sell-side analysts update EPS estimates promptly but rarely revise dividend forecasts with the same urgency. Income-oriented investors anchor on dividend history and are reluctant to reprice until the dividend change is formally announced. This creates a window of predictability: after a sufficiently negative earnings surprise, the probability of a dividend cut rises sharply, yet the stock often does not fully reflect this until the cut is announced weeks or months later. The reverse applies for positive surprises and dividend increases.

The clientele rotation mechanism. Unlike earnings-driven repricing, which is fast and sentiment-driven, dividend changes trigger a slower structural rotation in the investor base. After a dividend increase, the stock becomes more attractive to income-seeking investors (pension funds, insurance companies, retail retirees), whose buying pressure unfolds over weeks. After a cut, forced selling by income mandates and index rebalancing can suppress the stock for an extended period. This clientele-driven drift is what gives the

strategy a longer holding horizon than pure post-earnings momentum, potentially solving the turnover problem that plagues short-term event-driven strategies.

An important caveat: dividends are no longer the whole payout story. Since the early 2000s, share repurchases (buybacks) have increasingly replaced dividends as the dominant channel for returning cash to shareholders (Grullon & Michaely 2002b, Brav et al. 2005). A company that holds its dividend flat while substantially increasing its buyback programme is not necessarily signalling stagnation — it may simply prefer the flexibility and tax efficiency of repurchases. This means that a model focused exclusively on dividend changes will miss an important dimension of payout policy. You should be aware of this limitation throughout the project. In the feature engineering phase (Task 1.3), you will construct buyback-related features to help your model distinguish between genuine dividend stagnation and substitution. In the discussion section of your report, you should assess how much of your signal, if any, is contaminated by this substitution effect.

Your objective is to build an end-to-end pipeline that: (i) establishes the empirical existence of post-dividend-change return drift; (ii) identifies the conditions under which this drift is strongest; (iii) constructs a machine learning model to predict dividend changes using earnings-related features; and (iv) translates these predictions into a dollar-neutral long-short equity strategy with realistic performance evaluation.

1.1 The Earnings and Dividend Calendar on US Listed Stocks

Understanding the precise sequencing of corporate events is essential to this project. The diagram below maps the full calendar of earnings and dividend events for a typical US-listed stock. Every design decision in your strategy — when to enter, what to predict, how long to hold — follows from these dates.

Quarterly earnings, but not quarterly dividend decisions

US public companies are required by the SEC to report earnings every quarter. The vast majority of dividend-paying companies also *declare* a dividend every quarter. However, this does not mean that the dividend *amount* is reviewed every quarter. The typical pattern for a large-cap company is as follows: the board reviews the dividend level once a year, usually at a specific board meeting (often in Q1 or Q4, depending on the company's fiscal year), raises or cuts the dividend at that point, and then holds the per-share amount unchanged for the subsequent three quarters. A company paying $0.50 per share might raise to $0.55 in February and then declare $0.55 in each of May, August, and November with no further deliberation.

This distinction is critical for your model. A "no change" declaration in the company's annual review quarter is informative — the board actively chose not to raise despite having the opportunity. A "no change" in an off-quarter is expected and carries no signal

whatsoever. Your pipeline must identify each company's historical "review quarter" (the quarter in which changes have historically occurred) and weight those observations accordingly. You may wish to exclude off-quarter declarations from your training data entirely, or to create a binary feature flagging whether a given declaration falls in the review quarter.

Note that not all companies follow this annual-review pattern rigidly. Some companies (particularly REITs and utilities) adjust their dividend more frequently; a few pay monthly rather than quarterly; some pay semi-annually or annually; and special dividends can occur at any time. Your exploratory analysis (Task 1.4) should document the distribution of dividend frequencies and review-quarter patterns in your universe.

The five key dates in the dividend cycle

Each dividend event involves five dates, shown in the diagram below. Understanding which date carries information and which is purely mechanical is essential for avoiding subtle errors in your event study and strategy:

1. Earnings announcement. The company reports its quarterly results. This is the first event in the chain and the point at which your prediction model produces its signal. The earnings release contains the raw material (earnings surprise, guidance, cash flow data) that will inform the board's subsequent dividend decision. In this project, this is the earliest possible entry point (Entry A).

2. Dividend declaration (announcement) date. The board of directors formally announces the dividend amount, ex-date, record date, and payment date. This is the *information event* — the moment the market learns whether the dividend has changed. It typically occurs 2–4 weeks after the earnings release, often at the same board meeting or shortly thereafter. For your event study, this is day 0 when measuring post-announcement drift. It is also the second possible entry point (Entry B): entering after the declaration to ride the subsequent clientele-driven drift.

3. Ex-dividend date. The first trading day on which new buyers of the stock will *not* receive the upcoming dividend. The stock price typically opens lower by approximately the dividend amount on this date. Since May 28, 2024, US equity settlement operates on a T+1 basis, meaning the ex-dividend date and the record date are now typically the same business day (previously, under T+2, the ex-date was set one business day before the record date). To receive the dividend, an investor must purchase the stock at least one business day before the ex-date, so that the trade settles in time. The ex-date is set by the exchange, not by the company. For this project, the ex-date is primarily relevant to the mechanical price adjustment; it carries no new information beyond what was already revealed at the declaration.

4. Record date. The date on which the company's registrar examines the shareholder register to determine who is entitled to receive the dividend. Under T+1 settlement (effective May 28, 2024), the record date and the ex-dividend date are typically the same business day. If the record date falls on a non-business day, the ex-date is set to the preceding business day. This is a purely administrative date with no informational content.

5. Payment date. The date on which cash is actually transferred to shareholders' accounts. This is typically 2–4 weeks after the record date. It has no effect on the stock price and is irrelevant to your strategy.

Diagram: the full event calendar

The following diagram illustrates the annual quarterly cycle, the single-quarter event sequence, the corporate decision process, ex-date settlement mechanics under T+1, and the two strategy entry windows you will test.

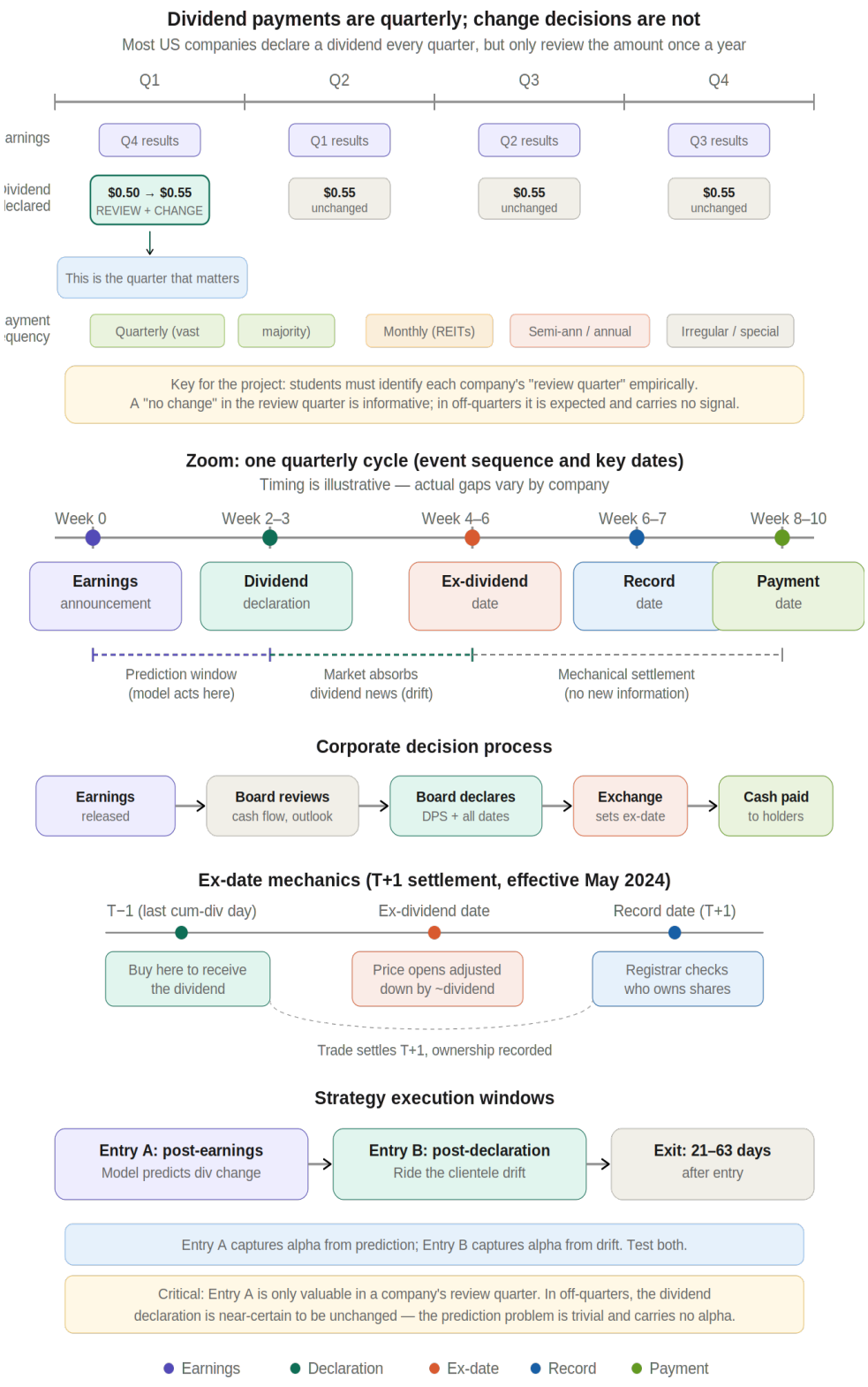

Figure 1. The earnings and dividend event calendar for US-listed equities, showing the quarterly payment cycle, the annual review pattern, single-quarter event sequence, settlement mechanics, and strategy entry windows.

Implications for strategy design

The calendar above implies three concrete constraints on your strategy. First, the **prediction window** (between the earnings release and the dividend declaration) is your opportunity to trade on information the market has not yet priced. The length of this window varies by company — some boards declare the dividend on the same day as earnings, others wait several weeks. You should measure this gap empirically in your data and treat it as a feature. Second, the **drift window** (after the declaration, before the ex-date and beyond) is where the clientele rotation occurs. Your event study in Phase 2 will measure how long this drift persists. Third, the **review quarter identification** is a prerequisite for your prediction model. A model that treats every quarterly declaration as an equal opportunity for a change will dilute its signal with thousands of uninformative "no change" observations from off-quarters. You must filter or weight appropriately.

Box 1 — S&P 500 Dividend Activity: What to Expect in Your Data

The following figures, drawn from S&P Dow Jones Indices quarterly reports (Howard Silverblatt, Senior Index Analyst), characterise the dividend landscape you will encounter. They are provided as orientation — your Task 1.4 exploratory analysis should reproduce and refine these numbers for your specific universe and sample period.

How many S&P 500 companies pay dividends?

As of Q4 2024, 405 issues (80.9%) of the S&P 500 pay a dividend (source: S&P DJI Q4 2024 report). This share has been remarkably stable: 403 (80.1%) at end-2023, 399 (79.5%) at end-2022. The remaining ~95–100 non-payers are concentrated in Technology and Healthcare (growth-oriented firms that prefer buybacks or reinvestment).

How often do they change?

In a typical expansion year, the large majority of payers raise their dividend at least once per calendar year. In 2024, S&P DJI reported $71.4 billion in total US common dividend increases vs. $18.1 billion in decreases, with a net increase of $53.3 billion (source: S&P DJI Q4 2024 report). The financial sector led with 61 increase events and industrials with 64. Four notable initiations in H1 2024 — Meta Platforms, Alphabet, Salesforce, and Bookings — added $16.4 billion in annualised dividend rate, illustrating that even large tech firms are entering the dividend-paying universe.

How rare are cuts?

Cuts and suspensions are very rare in normal times — but they spike dramatically in crises. In 2020, 43 S&P 500 issues suspended their dividends, producing a negative net change of –$40.8 billion (source: S&P DJI). This is the extreme class imbalance your model must handle: in benign years, the cut class may represent only 3–5% of payers, while in crisis years it can exceed 10%. The 2008–2009 financial crisis produced a similar spike. This asymmetry is precisely what Hypothesis H4 predicts and what your event study in Phase 2 will quantify.

Why this matters for your pipeline

These stylised facts confirm the design principles embedded in the coursework: (1) most payers review and adjust once per year in their review quarter, so off-quarter declarations are mechanical and carry no signal; (2) cuts are rare events concentrated in crises, creating severe class imbalance that your model must address explicitly (see Section 6.1); (3) sector

patterns vary substantially — Tech and Healthcare have fewer dividend changes (rarer = potentially stronger signal), while Financials, Utilities, and REITs adjust more frequently; (4) the entry of major tech firms into the dividend-paying universe (Meta, Alphabet, Salesforce in 2024) means the composition of your sample is shifting over time. Your Task 1.2 (dividend event classification) and Task 1.4 (summary statistics per year and sector) will reveal these patterns directly in your data.

2. Research Hypotheses

Your project should be guided by the following testable hypotheses. You are expected to formulate these precisely, state them formally where appropriate, and provide empirical evidence for or against each.

2.1 Primary Hypotheses

H1 (Post-Dividend-Change Drift): Stocks that experience a dividend increase (resp. decrease) exhibit positive (resp. negative) cumulative abnormal returns over the 5 to 63 trading days following the dividend announcement, after controlling for market and sector returns. Prior evidence from Michaely, Thaler & Womack (1995) documents +7.5% drift for initiations and −11.0% drift for omissions over the 12 months following the announcement, suggesting that the market under-reacts to dividend changes.

H2 (Earnings-Dividend Information Channel): The magnitude and direction of the most recent earnings surprise is a statistically significant predictor of the subsequent dividend change, even after controlling for standard fundamental variables (payout ratio, leverage, free cash flow coverage).

H3 (Conditional Drift Strength): The post-dividend-change drift is stronger when (a) the dividend change was not widely anticipated by the market, (b) the economic regime amplifies the signal (e.g., rate-tightening cycles for cuts), and (c) the stock belongs to a sector where dividend changes are rarer and therefore more informative.

2.2 Secondary Hypotheses

H4 (Asymmetric Drift): The negative drift following dividend cuts is larger in magnitude and longer in duration than the positive drift following dividend increases, due to asymmetric investor behaviour (loss aversion, forced selling by income mandates). Michaely, Thaler & Womack (1995) find that the post-omission drift (−11.0%) substantially exceeds the post-initiation drift (+7.5%), and that this asymmetry is distinct from and more pronounced than post-earnings-announcement drift.

H5 (Earnings Conditioning Improves Predictability): Among all dividend increases, stocks where the preceding earnings were also a positive surprise exhibit stronger post-announcement drift than those where earnings were in line or negative. This suggests that "genuine" dividend increases (backed by earnings improvement) carry more signal than "cosmetic" ones.

H6 (Machine Learning Outperforms Linear Models): A non-linear model (gradient-boosted trees or similar) produces a higher Information Coefficient for dividend change prediction than a ridge regression baseline, due to the presence of interaction effects between earnings features, leverage, and macro conditions.

3. Data Requirements

This section specifies the minimum data you will need. You are free to add additional data sources if justified.

3.1 Core Datasets

Dataset	Key Fields	Source Suggestions	Frequency
Dividend history	instrument_id, declaration_date, ex_date, record_date, payment_date, dividend_per_share, dividend_type (regular / special / initiation)	CRSP, Compustat, Yahoo Finance, EOD Historical Data	Event-level
Earnings announcements	instrument_id, announcement_date, actual_EPS, consensus_EPS, surprise (actual – consensus), pre_announcement_flag	IBES, Compustat, Estimize, provided class dataset	Event-level
Daily price data	instrument_id, date, adjusted_close, volume, market_cap	CRSP, all_prices.parquet (provided), Yahoo Finance	Daily
Fundamental data	instrument_id, date, ROA, net_cash_flow_oper,	Compustat, SEC EDGAR	Quarterly

	total_assets, long_term_debt, net_debt_to_equity, current_ratio, payout_ratio (or computable from EPS + DPS)		
Market / macro data	date, VIX, 10Y yield, 2Y yield (for yield curve slope), Fed Funds rate, S&P 500 return	FRED, Yahoo Finance	Daily
Sector classification	instrument_id, gics_sector, gics_industry_group, gics_industry	Compustat, provided class dataset	Static / annual

3.2 Universe Definition

Your universe should consist of S&P 1500 constituents (or a comparable broad US equity universe). You must clearly document:

- How you handle index membership changes (survivorship bias).
- Whether you apply any liquidity filters (minimum market cap, minimum average daily volume).
- How you treat stocks that enter or exit the universe during your sample period.

Minimum sample period: 2005–2024 (at least 15 years) to ensure coverage of multiple economic regimes including the 2008 financial crisis, the 2010–2019 expansion, the 2020 COVID shock, and the 2022–2023 rate-tightening cycle.

3.3 Data Integrity Requirements

- **No lookahead bias:** All features must be computable using only information available at the time of prediction. Dividend announcement dates, not ex-dates, are the relevant event dates for measuring when information becomes public.
- **Point-in-time fundamentals:** Accounting data must be lagged appropriately (typically one quarter) to reflect reporting delays.
- **Split/dividend adjustment:** Ensure price and dividend-per-share series are consistently adjusted for stock splits.
- **Missing data documentation:** Report the proportion of missing values for each variable and justify your imputation strategy.

- **Dividend-cut survivorship bias:** Companies that cut their dividend frequently delist thereafter (through bankruptcy, acquisition, or going private). This means your historical sample of dividend cuts is biased toward survivors — the worst outcomes have been removed. The post-cut drift you measure may therefore underestimate the true drift. Document how many dividend-cutting firms delist within 12 months of the cut, and discuss the implications for your backtest results.

4. Phase 1 — Data Preparation and Exploratory Analysis

Objective: Construct a clean, merged research dataset and perform exploratory analysis to understand the distributional properties of dividend events in your universe.

4.1 Tasks

Task 1.1: Data ingestion and cleaning

- Load all datasets, harmonise identifiers (instrument_id, date), and merge into a single research panel.
- Handle missing values: document your imputation strategy (forward-fill fundamentals within each stock, sector-median backfill, or MICE; justify your choice).
- Apply Winsorization to extreme values in fundamentals using expanding-window percentiles (1st and 99th) to avoid lookahead.
- Apply appropriate transformations: signed log for heavy-tailed accounting variables, z-score standardisation within each cross-section (date) for features that will enter the model.

Task 1.2: Dividend event classification

Create a classification of every dividend event in your universe:

- **Increase:** DPS is strictly higher than the previous regular dividend for the same stock.
- **Decrease:** DPS is strictly lower (including omission, coded as DPS = 0).
- **Unchanged:** DPS is identical to the previous regular dividend.
- **Initiation:** First-ever regular dividend payment for the stock in your sample.
- **Special dividend:** Non-recurring dividend. Flag these separately; they should be analysed but excluded from the main classification.

Compute the percentage change in DPS for each event. Exclude special dividends from the change calculation.

Task 1.3: Feature engineering

Construct the following features for each stock at each earnings announcement date. All features must be computable using only past information.

Feature Category	Features to Construct	Rationale
Earnings surprise	SUE (standardised unexpected earnings), earnings surprise magnitude and sign, analyst forecast dispersion, earnings revision momentum (change in consensus over prior 30/60/90 days)	Direct predictors of earnings quality and direction
Payout fundamentals	Payout ratio (DPS / EPS), payout ratio trajectory (change over 1/2/4 quarters), free cash flow coverage of dividends (FCF / total dividend payments), retained earnings ratio	Measures of dividend sustainability
Leverage and liquidity	Net debt-to-equity, current ratio, interest coverage ratio, debt maturity profile (if available), change in leverage over 4 quarters	Constraints on dividend capacity
Historical dividend behaviour	Number of consecutive quarters of dividend growth, years since last dividend cut, historical dividend growth rate (1Y, 3Y), coefficient of variation of past DPS	Management's revealed preference for dividend smoothing
Market and sentiment	Short interest ratio, days since last corporate news (dtcn), VIX level, 10Y–2Y yield spread, recent stock return momentum (1M, 3M, 6M), stock volatility (1M, 3M)	Market-based risk and sentiment signals
Sector context	Sector average dividend yield, sector average payout ratio, number of dividend changes in the same sector in the prior quarter, sector-relative earnings surprise rank	Peer-group conditioning
Buyback and total payout	Net buyback yield (share repurchases – issuances, scaled by market cap), total payout yield (dividend yield + net	Distinguishes genuine dividend stagnation from substitution toward

	buyback yield), change in buyback intensity over 1/4 quarters, buyback-to-dividend ratio	buybacks (Grullon & Michaely 2002b)

Task 1.4: Exploratory analysis

Produce the following analyses and include them in your report:

1. Summary statistics of dividend events: number of increases, decreases, omissions, initiations per year, broken down by GICS sector.
2. Distribution of dividend change magnitudes (histogram, by direction).
3. Cross-tabulation of earnings surprise quintile vs. subsequent dividend action (increase / maintain / cut). Is there a visible pattern?
4. Correlation matrix of your engineered features. Identify and comment on any high-correlation pairs.
5. Time-series plot of dividend cut frequency overlaid with VIX and yield curve slope. Discuss any regime-dependent patterns.

5. Phase 2 — Event Study: Post-Dividend-Change Return Dynamics

Objective: Establish whether a tradeable post-dividend-change drift exists, and characterise its magnitude, duration, and asymmetry. This phase requires no machine learning — it is a pure empirical analysis.

5.1 Tasks

Task 2.1: Cumulative Abnormal Return (CAR) computation

For each dividend announcement in your sample, compute:

- **Raw return:** Cumulative stock return over windows [0, +T] where T ∈ {1, 2, 3, 5, 10, 21, 42, 63, 126} trading days after the announcement.
- **Market-adjusted return:** Subtract the cumulative S&P 500 return over the same window.
- **Sector-adjusted return:** Subtract the cumulative GICS sector ETF (or equal-weighted sector portfolio) return over the same window.

The event date (day 0) is the dividend declaration date, not the ex-date. If the declaration occurs after market close, day 0 is the next trading day.

Task 2.2: Aggregate CAR analysis

Compute average CARs separately for:

1. All dividend increases
2. All dividend decreases (including omissions)
3. Dividend initiations
4. All unchanged dividends (this is your control group)

For each group, report the mean CAR, median CAR, standard error, t-statistic, and the number of events. Plot the average CAR path from day 0 to day +126 with confidence bands. This is the most important chart in your entire project.

Task 2.3: Asymmetry analysis (H4)

Test whether the negative drift following cuts is larger and/or more persistent than the positive drift following increases. Perform a formal statistical test (e.g., Welch's t-test on CARs at each horizon, or a bootstrap test). Discuss the economic implications.

Task 2.4: Conditional drift analysis (H3)

Slice your CAR analysis along the following conditioning dimensions. For each, produce a comparison plot showing average CAR paths for the subgroups:

Conditioning Dimension	Subgroups	Expected Pattern
Dividend change magnitude	Small (≤10th pctile of |change|), Medium, Large (≥90th pctile)	Larger changes should produce stronger drift
GICS sector	At minimum: Financials, Technology, Utilities, Healthcare, Consumer Discretionary	Rarer dividend events (Tech) should carry more signal
VIX regime	Low (<15), Medium (15–25), High (>25)	Drift may be longer in low-vol regimes
Rate cycle	Tightening (rising Fed Funds rate), Easing, Stable	Cuts during tightening may signal deeper trouble

Earnings surprise interaction	Positive surprise + dividend increase; Negative surprise + dividend cut; Contradictory (e.g., positive earnings but dividend cut)	Aligned signals should produce strongest drift; contradictory signals are ambiguous
Market cap	Small (bottom 30%), Mid (30–70%), Large (top 30%)	Smaller stocks may have less analyst coverage and longer drift

Task 2.5: Statistical rigour

- Control for multiple testing (Bonferroni or Holm correction) when comparing many subgroups.
- Report bootstrapped confidence intervals (at least 1,000 iterations) for key CAR estimates.
- Discuss potential confounds: are dividend changes clustered in time? Do cuts coincide with market-wide drawdowns that inflate your CARs?

6. Phase 3 — Dividend Change Prediction Model

Objective: Build a machine learning model that, at each earnings announcement, predicts the probability that the stock's next dividend will be an increase, a decrease, or unchanged. This model will generate the signals for your trading strategy.

6.1 Target Variable

The target variable is the dividend action that occurs at the next dividend declaration after the earnings announcement:

- **y = +1** (increase): DPS increases relative to the prior regular dividend.
- **y = 0** (unchanged): DPS is identical.
- **y = −1** (decrease/omission): DPS decreases or is omitted.

Critical timing constraint: The target is the dividend decision that has NOT yet been announced at the time of the earnings release. You must verify that for every observation in your dataset, the next dividend declaration date is strictly after the earnings announcement date. Any violation of this constraint constitutes lookahead bias and will be penalised.

Class imbalance warning: The target distribution will be heavily imbalanced. In a typical quarter, 85–90% of declarations are "unchanged," with increases and decreases each

representing a small minority. If you only consider review-quarter declarations (as recommended in Section 1.1), the imbalance shifts but remains significant — increases may represent 30–40% of review-quarter observations, but decreases remain rare (typically under 5% outside of recessions). You must address this imbalance explicitly. Options include: using class weights in the loss function (e.g., inverse-frequency weighting), framing the problem as two separate binary classifiers (P(increase) and P(decrease)), using stratified sampling in your walk-forward folds, or evaluating with metrics that are robust to imbalance (AUC-ROC, precision-recall curves) rather than accuracy alone. Additionally, a model that treats "unchanged" dividend declarations from off-quarters as equal training observations will be overwhelmed by trivially predictable non-events. Consider restricting your training data to review-quarter observations or, at minimum, heavily downweighting off-quarter samples.

Buyback substitution and the "unchanged" label: Be aware that y = 0 ("unchanged") is not a homogeneous category. A company that holds its dividend flat while dramatically increasing its share repurchase programme is economically very different from one that holds its dividend flat and does nothing. Your buyback features (Task 1.3) can help the model distinguish between these cases. You may also wish to experiment with a more granular target that incorporates total payout changes, though this adds complexity and should be treated as an extension rather than the baseline.

6.2 Model Architecture

Task 3.1: Baseline model — Ridge regression

Implement a Ridge regression (or logistic regression with L2 penalty) as your baseline. This model should use all features from Task 1.3. Report the Information Coefficient (Spearman rank correlation between predicted probabilities and realised outcomes) and the confusion matrix.

Task 3.2: Primary model — Gradient-boosted trees

Implement a LightGBM (or XGBoost / CatBoost) classifier. You may frame this as:

- **Three-class classification** (increase / unchanged / decrease), or
- **Two separate binary models** (P(increase) and P(decrease)), whose outputs are combined into a composite score: score = P(increase) – P(decrease). The "two-model" approach may be more effective if the features that predict increases differ from those that predict cuts.

Justify your choice of framing and hyperparameters.

Task 3.3: Advanced model

Implement one or more of the following:

- A two-stage architecture where Stage 1 predicts an auxiliary target (e.g., the earnings surprise itself), and Stage 2 uses the Stage 1 out-of-fold predictions as a feature for dividend prediction.
- An FT-Transformer or TabNet model, with a discussion of whether the self-attention mechanism captures interaction effects that tree models miss.
- NLP features from earnings call transcripts (e.g., frequency of phrases like “capital allocation,” “payout flexibility,” “preserving cash”) added as features to the tree model.

6.3 Validation Framework

Walk-forward validation is mandatory. You must use an expanding or rolling window that respects temporal ordering. At minimum, use 5 folds with training windows of at least 5 years. No information from the validation or test period may leak into training, including through feature engineering, target encoding, or imputation.

For each fold, report:

1. Information Coefficient (Spearman rank correlation).
2. AUC-ROC for the binary sub-models (if using the two-model approach).
3. Confusion matrix and precision/recall for each class.
4. Feature importance (SHAP values for tree models, coefficients for Ridge).

Additionally, report IC stability across folds. If the IC is positive in some folds and negative in others, discuss what this means for the robustness of the signal.

6.4 Model Diagnostics

You are expected to provide:

- A SHAP summary plot showing the top 15 features and their directional impact.
- A partial dependence plot for the 3 most important features.
- An analysis of prediction errors: in which sectors or regimes does the model fail most? Are false positives (predicted increase, actual cut) concentrated in particular conditions?

- A calibration plot: are predicted probabilities well-calibrated?

7. Phase 4 — Trading Strategy Construction

Objective: Translate the dividend change predictions from Phase 3 into a dollar-neutral long-short equity strategy and evaluate its performance under realistic assumptions.

7.1 Strategy Design

Task 4.1: Signal construction

At each earnings announcement, your model produces a score for the stock. Define:

- **Long candidates:** Stocks with high P(dividend increase) and low P(dividend decrease). These are firms where the earnings signal and fundamental context point to an imminent dividend hike and a subsequent positive drift.
- **Short candidates:** Stocks with high P(dividend decrease). These are firms where the model predicts a dividend cut and a subsequent negative drift.

You must define clear, rule-based thresholds for entry (e.g., score in the top / bottom quintile of the cross-sectional distribution on that date). Thresholds must be computed using only past data (expanding-window quantiles).

Task 4.2: Portfolio construction

The strategy must be dollar-neutral: at every rebalancing date, the total long exposure (in dollars) must equal the total short exposure. Specifically:

- **Entry:** A position is opened on the trading day after the earnings announcement (to avoid using closing-price information from the announcement day). If you choose to also condition on the dividend announcement, document the alternative entry point.
- **Weighting:** You must implement and compare at least two weighting schemes: (a) Equal-weight within each leg. (b) Score-weighted (position size proportional to |score|, normalised so that each leg sums to the same dollar amount).
- **Holding period:** Test holding periods of 5, 10, 21, 42, and 63 trading days. The Phase 2 event study should inform your choice of optimal holding period. Positions are exited at the close on the last day of the holding period.

- **Dollar neutrality:** At every rebalancing date, ensure Σ(long weights) = Σ(|short weights|). If on a given date there are insufficient long or short candidates, scale down both legs proportionally.

Task 4.3: Rebalancing mechanics

Given that earnings announcements are event-driven (not on a fixed calendar), you need to handle overlapping positions carefully:

- If a new signal arrives for a stock that already has an open position, describe your rule (e.g., extend the holding period, or close and re-enter, or ignore the new signal).
- Positions from different entry dates will coexist. Document how you aggregate positions and maintain dollar neutrality across overlapping cohorts.

7.2 Transaction Cost Model

You must incorporate realistic transaction costs:

Cost Component	Specification
Commission	Assume 0 (most institutional brokers charge zero commission on US equities)
Market impact / half-spread	5 bps per trade for large-cap stocks (S&P 500 constituents); 10 bps for mid/small-cap
Short borrowing cost	Assume 50 bps annualised for general collateral names; flag any hard-to-borrow names in your universe
Slippage	Assume execution at the volume-weighted average price (VWAP) of the first 30 minutes after market open, approximated as the midpoint between open and close

Perform a breakeven analysis: at what transaction cost level does the strategy's Sharpe ratio fall below 1.0? Below 0.5? Below the market benchmark?

7.3 Risk Management

- **Position limits:** No single stock may exceed 5% of the total portfolio on either leg.
- **Sector concentration:** No single GICS sector may exceed 30% of either leg.

- **Drawdown control:** Describe (but you are not required to implement) a mechanism that would reduce position sizes after a drawdown exceeding 15%.

8. Phase 5 — Performance Evaluation and Reporting

Objective: Evaluate the strategy rigorously and produce a professional-quality research report.

8.1 Performance Metrics

Report the following for each strategy variant (weighting scheme × holding period):

Metric	Definition / Notes
Annualised return (CAGR)	Geometric mean annual return
Annualised volatility	Standard deviation of daily returns × $\sqrt{252}$
Sharpe ratio	(CAGR – risk-free rate) / annualised volatility; use 3-month T-bill as risk-free rate
Maximum drawdown	Largest peak-to-trough decline in cumulative return
Calmar ratio	CAGR / |maximum drawdown|
Average turnover	Average fraction of portfolio replaced at each rebalancing
Hit rate	Fraction of trades with positive return
Profit factor	Gross profits / gross losses
Long leg CAGR	Performance of the long book in isolation
Short leg CAGR	Performance of the short book in isolation (should be negative for a profitable short book)
Information Coefficient (strategy-level)	Rank correlation between model score and realised forward return across all observations

8.2 Attribution and Robustness

Task 5.1: Factor regression

Regress your strategy's daily returns against the Fama-French 5-factor model (Mkt-RF, SMB, HML, RMW, CMA) plus momentum (UMD). Report:

- The alpha (intercept) and its t-statistic.

- Factor loadings. Is your strategy exposed to Value (HML)? To Momentum (UMD)? If so, how much of your return is explained by known factors?
- The adjusted R-squared. A low R-squared with a significant alpha is the ideal result — it means your signal is not a repackaging of known factors.

Task 5.2: Regime analysis

Split your backtest into sub-periods and report performance for each:

1. Pre-2008 (if your data goes back far enough).
2. 2008–2009 (financial crisis).
3. 2010–2019 (expansion and low-rate environment).
4. 2020 (COVID crash and recovery).
5. 2021–2022 (rate-tightening cycle).
6. 2023–2024 (recent).

Does the strategy work consistently, or is performance concentrated in one regime? This is a critical credibility check.

Task 5.3: Robustness checks

- **Sector exclusion:** Re-run the strategy excluding Utilities and REITs (sectors where dividend changes are most frequent and potentially least informative). Does the signal survive?
- **Model simplification:** Replace the ML model with a simple rule-based strategy (e.g., “go long stocks with positive earnings surprise and payout ratio < 50%; go short stocks with negative earnings surprise and payout ratio > 80%”). How much alpha does the ML model add over a naive rule?
- **Holding period sensitivity:** Produce a heatmap of Sharpe ratio as a function of holding period (x-axis) and entry threshold (y-axis). Is there a stable “sweet spot,” or is performance sensitive to these parameters?
- **Out-of-sample test:** Reserve the final 2 years of data as a true out-of-sample test. The model must not be retrained or tuned on this period. Report all metrics on this holdout.

9. Technical Specifications

9.1 Code Organisation

Your codebase must be modular and reproducible. The following file structure is recommended:

File	Purpose
config.py	All configurable parameters: file paths, universe filters, feature lists, model hyperparameters, holding periods, cost assumptions
data_loader.py	Data ingestion, merging, cleaning, missing value handling
feature_engineering.py	All feature construction. Must contain only pure functions: def compute_features(df: pd.DataFrame) -> pd.DataFrame
target_engineering.py	Dividend event classification and target variable creation
event_study.py	CAR computation, conditional drift analysis, plotting (Phase 2)
model.py	Model training, walk-forward validation, prediction, SHAP analysis (Phase 3)
strategy.py	Signal construction, portfolio weighting, dollar-neutral construction, holding period management (Phase 4)
backtest.py	Return computation, transaction cost application, performance metrics, factor regression (Phase 5)
utils.py	Shared helper functions (e.g., expanding-window quantile, Winsorization, signed log transform)
main.py	End-to-end pipeline runner: data → features → model → strategy → backtest

9.2 Coding Standards

- **Language:** Python only. Use pandas DataFrames and numpy arrays. Do not use R, MATLAB, or Excel.
- **No for-loops on dates:** Use groupby operations wherever possible. Loops on individual dates are extremely slow for panel data and will be penalised unless justified and optimised (e.g., with numba).
- **Reproducibility:** Fix all random seeds. The pipeline must produce identical results when re-run.
- **Output format:** All intermediate and final datasets should be saved as .parquet files (not .csv) for type safety and compression.

- **Functions, not scripts:** All logic must be encapsulated in well-documented functions with type hints. No code should execute at module-level except in main.py.

9.3 Anti-Lookahead Checklist

Before submission, verify each of the following. Any violation will result in a significant mark deduction:

Check	Question to Ask Yourself
Feature computation	At time t, does every feature value depend only on information available before t?
Winsorization thresholds	Are clip percentiles computed on an expanding window of past data, not the full sample?
Standardisation parameters	Are mean and standard deviation fitted on training data only, then applied to validation?
Target variable	Is the dividend change label based on a declaration date strictly after the feature observation date?
Model hyperparameters	Were hyperparameters tuned only on training/validation folds, never touching the final holdout?
Cross-sectional ranks	Are decile/quintile thresholds computed on the current date's cross-section, not the full panel?
Transaction execution	Are trades executed at prices available after the signal date (e.g., next-day open), not at the signal-day close?

10. Deliverables

10.1 Written Report (max 20 pages, excluding appendices)

The report should follow the structure of a quantitative research paper:

1. **Abstract** (200 words): state the research question, method, and key finding.
2. **Introduction:** motivation, literature review (at least 8 academic references), and hypotheses.
3. **Data and Feature Engineering:** describe your data sources, universe, features, and transformations.

4. **Event Study Results:** Phase 2 findings. This must come before the model, as it establishes the economic foundation.
5. **Model:** architecture, validation, feature importance, diagnostics.
6. **Strategy and Backtest:** portfolio construction, performance, factor attribution, robustness.
7. **Discussion:** what works, what doesn't, limitations, and suggestions for future work.
8. **Conclusion:** answer the research question directly. Is there a tradeable, risk-adjusted alpha in the earnings-to-dividend information channel?

10.2 Code Submission

- Complete, runnable Python codebase following the structure in Section 9.1.
- A README.md with setup instructions, dependencies, and a single command to reproduce all results.
- A requirements.txt or environment.yml specifying exact package versions.

11. Marking Criteria

Component	Weight	What Earns Top Marks
Data handling and feature engineering	15%	Clean, well-documented pipeline. Creative and well-motivated features. Robust imputation. No lookahead violations.
Event study (Phase 2)	15%	Rigorous statistical analysis. Clear visualisations. Thoughtful conditional analysis. Proper controls for multiple testing.
Dividend prediction model (Phase 3)	20%	Appropriate model selection with justification. Walk-forward validation. SHAP analysis. Calibration. Clear comparison between Ridge baseline and ML model.
Trading strategy and backtest (Phase 4–5)	25%	Dollar-neutral construction. Realistic transaction costs. Fama-French factor regression. Regime analysis. Breakeven analysis. Robustness checks.
Report quality	15%	Clear writing. Professional formatting. Logical flow from hypothesis to evidence. Honest discussion of limitations.

Code quality	10%	Modular, documented, reproducible. Efficient use of pandas groupby (no unnecessary loops). Type hints. Clean separation of concerns.

Distinction-level work will demonstrate: (i) a genuine understanding of why the strategy works (or doesn't) rather than just reporting metrics; (ii) at least one creative extension beyond the minimum requirements (e.g., NLP features, regime-switching model, dynamic holding period); (iii) an honest assessment of whether the alpha would survive in a live trading environment.

Failing work will exhibit one or more of: undetected lookahead bias; no walk-forward validation; reporting backtest results without transaction costs; inability to explain model outputs; or plagiarism.

12. Suggested References

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Brav, A., Graham, J. R., Harvey, C. R., & Michaely, R. (2005). Payout Policy in the 21st Century. Journal of Financial Economics, 77(3), 483–527.

Appendix A: Mathematical Notation Reference

CAR(i, t, T) = cumulative abnormal return for stock i from day t to day t+T, where the abnormal return on each day is r(i,d) – r(benchmark, d).

SUE(i, q) = (EPS_actual – EPS_consensus) / σ(past forecast errors) for stock i in quarter q.

Payout ratio(i, q) = DPS(i, q) × 4 / EPS(i, trailing 4Q). Handle negative EPS by capping the ratio or using an indicator.

Dollar neutrality condition: Σ(w_i) for i in Long = Σ(|w_j|) for j in Short, at every rebalancing date.

IC = Spearman rank correlation between the model's predicted score and the realised forward return, computed cross-sectionally on each date and then averaged.

Appendix B: Timeline and Milestones

Week	Milestone	Deliverable
1–2	Data collection, cleaning, feature engineering	Clean merged dataset (.parquet), EDA notebook
3–4	Event study (Phase 2)	CAR analysis complete, key charts produced
5–6	Dividend prediction model (Phase 3)	Trained model, walk-forward IC results, SHAP plots

7–8	Strategy construction and backtest (Phase 4–5)	Full backtest results, factor regression, robustness
9–10	Report writing and presentation preparation	Final report, code submission, presentation slides